Global Weather and Air Quality Analysis (2024 – 2026) using Python

1.Project Overview

Global Weather and Air Quality Analysis is a Python-based data analytics project that explores global weather and air quality patterns across countries and cities. The project identifies **temperature and weather trends, analyses pollution levels and hotspots,** and examines **relationships between weather parameters and air quality indicators.**

2.Dataset Information

- **Source** : World data foundation.org
- **Dataset** : Global Weather & Air Quality Data
- **Dataset Link** : https://worlddatafoundation.org/datasets
- **Geographical Coverage :** Global (Multiple Countries & Cities)
- **Format** : CSV
- **Domain** : Weather & Air Quality
- **Timeline** : 2024 – 2026
- **Number of Records** : 1,25,696
- **Number of Attributes** : 41

3.Business Problem

Air quality and weather conditions vary significantly across different countries and cities, making it difficult to identify pollution hotspots and understand the environmental factors associated with poor air quality. The business problem is to analyze **global weather and air-quality** data to **identify high-pollution locations, understand patterns and relationships between weather conditions and air-quality indicators**, and provide data-driven insights that can support effective environmental monitoring, pollution management, and decision-making.

4.Objective

1. To analyze the distribution and variability of temperature across the dataset.
2. To identify and visualize the most frequently occurring weather conditions in the dataset.
3. To analyze global weather conditions across different countries.
4. To identify relationships between weather parameters and air quality indicators using correlation analysis.
5. To analyze the relationship between actual temperature and feels-like temperature.
6. To identify locations with higher pollution levels using air quality indicators and pollution categories.
7. To analyze the monthly distribution and variation of pollution categories.
8. To examine the relationship between temperature and PM2.5 levels across different pollution categories.
9. To identify hourly patterns in PM2.5 and PM10 pollution levels.

5.Column Description

Column Name	Data Type	Description
country	object	The name of the country where the weather and air-quality observation was recorded.
Location_name	object	The name of the city or specific location where the observation was recorded.
latitude	float64	The north–south geographic coordinate of the location, measured in degrees.
longitude	float64	The east–west geographic coordinate of the location, measured in degrees.
time/zone	object	The local time zone of the recorded location.
last_updated	object	The date and time when the weather and air-quality data was last updated.
temperature_celsius	float64	The actual air temperature at a specific location and time, measured in degrees Celsius (°C).
Weather_condition	object	A text description of the observed weather condition, such as **Sunny, Cloudy, or Rainy.**
wind_kph	float64	The wind speed recorded at the location, measured in kilometres per hour (km/h).
wind_degree	int64	The direction from which the wind is blowing, measured in degrees.
pressure_mb	int64	The atmospheric pressure at the location, measured in millibars (mb).
Precip_mm	float64	The amount of precipitation recorded at a specific location and time, measured in millimetres (mm).
Humidity	int64	The percentage of moisture present in the air at a specific location and time.
cloud	int64	The percentage of the sky covered by clouds.
feels_like_celsius	float64	The perceived temperature experienced by people, measured in degrees Celsius (°C).
visibility_km	float64	The distance at which objects can be clearly seen, measured in kilometres (km).
uv_index	float64	A measure of the intensity of ultraviolet (UV) radiation from the sun.
gust_kph	float64	The maximum or sudden increase in wind speed, measured in kilometres per hour (km/h).
air_quality_Carbon_Monoxide	float64	The concentration of carbon monoxide (CO) in the air, indicating a level of air pollution.
air_quality_Ozone	float64	The concentration of ozone (O_3) present in the air, used as an indicator of air quality.
air_quality_Nitrogen_dioxide	float64	The concentration of nitrogen dioxide (NO_2) in the air, an important air-pollution indicator.
air_quality_PM2.5	float64	The concentration of fine particulate matter with a diameter of 2.5 micrometres or less in the air.
air_quality_PM10	float64	The concentration of particulate matter with a diameter of 10 micrometres or less in the air.
air_quality_us-epa-index	int64	The US EPA Air Quality Index value used to indicate the overall level of air pollution and its potential health impact.

6.Tools & Technologies

- Language - Python
- Environment - Google colab
- Data Manipulation - pandas, Numpy
- Data Visualization – matplotlib, seaborn

7.Data Analytics Life Cycle

A **Data Analytics Pipeline** is a step-by-step process used to convert **raw data into meaningful insights**.

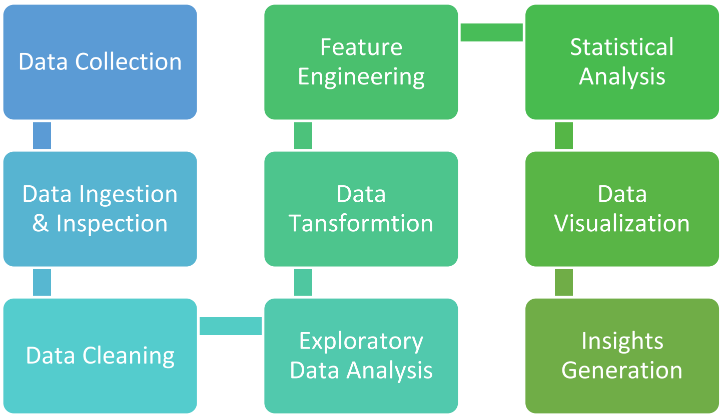


Figure 1: End-to-end analytics life cycle followed in this project

7.1 Data Ingestion

- The dataset was loaded from **Google Drive into Google Colab** using the **Pandas** library.
- Google Drive was mounted in the Colab environment, and the raw CSV dataset was imported for further data cleaning, preprocessing, and analysis.

Import Libraries

The required Python libraries were imported using the following code:

```
import pandas as pd
```

Data Ingestion from Google Drive

Google Drive was mounted to access the dataset stored in the user's Google Drive.

```
from google.colab import drive
drive.mount(/content/drive)
```

Load the Dataset

The raw Global Weather & Air Quality Analytics dataset was loaded into a Pandas **DataFrame** using the **read_csv()** function.

```
df = pd.read_csv(/content/drive/MyDrive/Python Capstone Poject/Global Weather & Air Quality Analytics Raw Data.csv)
```

7.2 Data Inspection

- After loading the dataset from Google Drive into Google Colab using Pandas, an initial data inspection was performed to understand the structure and quality of the raw dataset before proceeding with data cleaning and preprocessing.
- The following checks were performed:

  df.head() – To view the first five records of the dataset.
  df.tail() – To view the last five records of the dataset.
  df.shape – To identify the number of rows and columns.
  df.info() – To examine the column names, data types, and non-null counts.
  df.columns – To display all column names.
  df.describe() – To obtain the statistical summary of numerical variables.
  df.dtypes – To verify the data type of each column.

7.3 Data Cleaning & Preprocessing

During the data cleaning and preprocessing stage, the dataset was carefully reviewed to ensure data quality, consistency, and reliability before proceeding with further analysis.

Removal of Unwanted Columns:

- Based on the project objectives and analysis requirements, **unnecessary and redundant columns** were identified and removed from the dataset.
- Some columns contained the **same information in different units**, such as temperature in Celsius and Fahrenheit, wind speed in KPH and MPH, and visibility in kilometres and miles.
- To avoid duplication and **maintain consistency, only the required unit-based columns were retained.**
- In addition, columns that were not directly relevant to the project's weather and air quality analysis were removed. This included timestamp-related, moon-related, and certain air quality index columns.

- The **removed columns** include:

  last_updated_epoch, temperature_fahrenheit, wind_mph, pressure_in, precip_in, feels_like_fahrenheit, visibility_miles, gust_mph, air_quality_Sulphur_dioxide, air_quality_gb-defra-index, moonrise, moonset, moon_phase, moon_illumination, sunrise, and sunset.

This column reduction helped eliminate duplicate information, reduce dataset complexity, and create a more focused dataset for further analysis.

Remaining Columns:

- After removing the unwanted and redundant columns, the **shape of the dataset was checked** to verify the changes made during the column removal process.
- The remaining columns were then displayed and reviewed to confirm that all relevant variables required for the project were retained.

Handling Missing Values:

- The dataset was checked for missing values using the **isnull().sum()** method. The result showed that all selected columns contained **zero missing values**.
- Therefore, **no missing value imputation or replacement techniques were required**.

.**Duplicate Check:**

- Duplicate records were checked using the **duplicated().sum()** method after removing the unwanted and redundant columns.
- The result showed that the dataset contained **no duplicate records**, ensuring that each observation was unique and suitable for further analysis.

Standardization of Country Names:

- The unique values in the **country column** were first examined to identify inconsistent country names. During this inspection, some country names were found to be represented in **different languages, spellings, and naming** formats.
- To ensure consistency in country-wise grouping and analysis, a **mapping dictionary** was created to convert these inconsistent representations into standardized country names. The mapping was then applied to the country column.
- For example, **Malásia** was converted to **Malaysia**, **كولومبيا** to **Colombia, Гватемала** to **Guatemala**, and **火鸡** to **Turkey**.

Removal of Invalid Location Entry:

- The **location_name** column was examined for invalid or inconsistent entries. An invalid location entry named **#NAME?** was identified, which does not represent an actual geographical location. Therefore, the corresponding record was removed from the dataset to maintain data accuracy.

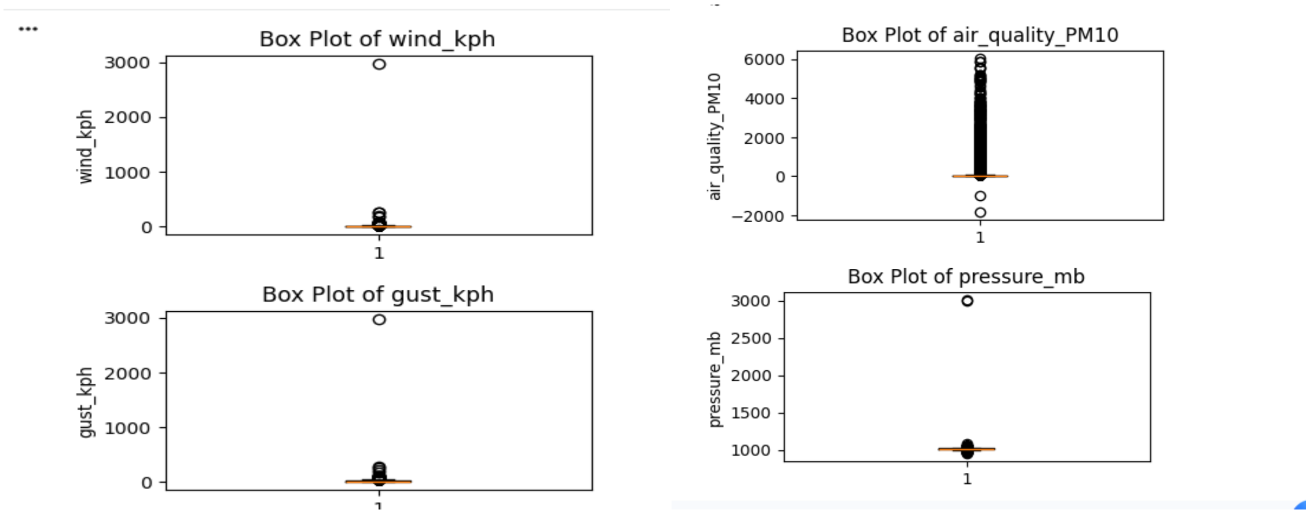
- Overall, these data cleaning steps improved the **consistency, accuracy, and reliability** of the dataset and prepared it for further exploratory data analysis and visualization.

Outlier Detection and Data Validation:

- As part of the data validation process, potential outliers were identified using **box plots** for selected numerical variables.
- This analysis helped detect unusually high or low values that could indicate **potential data errors or inconsistencies** in the dataset.
- The following variables were analyzed:

wind_kph, gust_kph, air_quality_Carbon_Monoxide,

air_quality_PM10, pressure_mb



The box plots helped identify extreme and potentially unrealistic values in the dataset. During the validation process, several **data errors** were identified:

- **Wind Speed:** An unrealistic value of **2963.2 km/h** was identified in the wind_kph column.
- **Gust Speed:** An unrealistic value of **2970.4 km/h** was identified in the gust_kph column.
- **Carbon Monoxide:** The air_quality_Carbon_Monoxide column contained a negative value of **-9999**, which is physically invalid.
- **PM10:** The air_quality_PM10 column contained a negative value of **-1848.15**, which is also physically invalid. A negative value of -1848.15 was identified... Therefore, **all negative** PM10 values were **removed.**
- **Atmospheric Pressure:** Extreme values of **3000 mb** and **3006 mb** were identified in the pressure_mb column, which are unrealistic for normal atmospheric conditions.

After removing the clearly invalid and unrealistic data errors identified during data validation, the remaining outliers were further examined using box plots and statistical analysis.

Some observations were identified as statistical outliers; however, they were not automatically removed because extreme weather and air-quality values can represent genuine environmental conditions.

Therefore, only values that were clearly identified as **physically invalid or data-entry errors** were removed. The remaining statistically identified outliers were retained to preserve the natural variation and extreme conditions present in the dataset.

This approach helps avoid unnecessary data loss and ensures that potentially meaningful extreme weather and air-quality observations are retained for further analysis.

7.4 Feature Engineering

During the feature engineering stage, three new features were created from the existing columns to support more meaningful analysis and visualization.

Pollution Category:

- A new column named **pollution_category** was created using the **air_quality_us-epa-index** column. The EPA air quality index values were mapped into meaningful categories such as **Good, Moderate, Unhealthy for Sensitive Groups, Very Unhealthy, and Hazardous**.
- This feature helps in analyzing and comparing air quality levels across different locations.

Temperature Category:

- A new column named **temperature_category** was created based on the **temperature_celsius** column. Temperature values were grouped into meaningful temperature ranges or categories such as **Freezing, Cold, Moderate, Hot.**
- This feature helps to understand the distribution of weather conditions and compare air quality across different temperature levels.

Month Name Extraction:

- The **month_name** feature was extracted from the **last_updated** datetime column to analyze monthly variations and seasonal patterns in temperature and air-quality levels.

Hour Extraction:

- A new column named **update_hour** was created by extracting the hour component from the **last_updated** datetime column.
- This feature enables hour-wise analysis and helps identify patterns or variations in weather and air quality throughout the day.

New Feature	Derived from	Purpose
Pollution_category	air_quality_us-epa-index	To classify and compare pollution levels across locations.
Temperature_category	temperature_celsius	To analyze and compare temperature conditions across locations.
Month_name	last_updated	To analyze monthly and seasonal variations
Update_hour	last_updated	To analyze hourly weather and pollution patterns

7.4 Data Transformation

As part of the data transformation process, the dataset was modified to improve data consistency, readability, and usability for further analysis.

Date-Time Conversion:

The last_updated column was initially stored as an **object (string) data type**. It was converted into the **datetime data type** to enable proper date- and time-based analysis, sorting, filtering, and visualization.

Column Name Standardization:

The column name **condition_text** was renamed to **weather_condition** to make the column name more meaningful, descriptive, and easier to understand during analysis.

7.5 Final EDA Validation

After completing the **data cleaning, data transformation, and feature engineering** processes, a final Exploratory Data Analysis (EDA) check was performed on the transformed dataset.

- Verified the **final number of rows(125691) and columns(29)** in the dataset.
- Reviewed the **remaining column names** and their data types.
- Rechecked for **missing values** to ensure that no unexpected null values were introduced during the transformation process.
- Rechecked for **duplicate records** to confirm data uniqueness.
- Validated the newly created **pollution category**, **temperature category**, and **hour** columns.
- Reviewed the numerical variables to ensure that previously identified **invalid and unrealistic values** had been properly handled.
- Checked the overall data structure and value distributions to identify any remaining inconsistencies.

7.6 Statistical Analysis

Measure of Central Tendency

measures of central tendency were calculated for the selected numerical variables. Mean, Median, and Mode were used to understand the typical values and distribution patterns of the weather and air-quality parameters.

The analysis included the following variables:

Measure of Central Tendency	Mean	Median	Mode
temperature_celsius	21.566100	24.100	26.30
wind_kph	12.959654	11.200	3.60
pressure_mb	1014.024425	1014.000	1013.00
precip_mm	0.136675	0.000	0.00
humidity	66.179440	71.000	94.00
cloud	40.036884	31.000	0.00
feels_like_celsius	22.455924	25.300	24.60
visibility_km	9.516108	10.000	10.00
uv_index	3.430491	2.100	0.00
gust_kph	18.289470	15.500	10.80
air_quality_Carbon_Monoxide	475.327953	302.850	217.00
air_quality_Ozone	58.849419	56.000	51.00
air_quality_Nitrogen_dioxide	15.234244	5.500	0.00
air_quality_PM2.5	24.558451	14.245	0.50
air_quality_PM10	49.175345	20.250	12.95

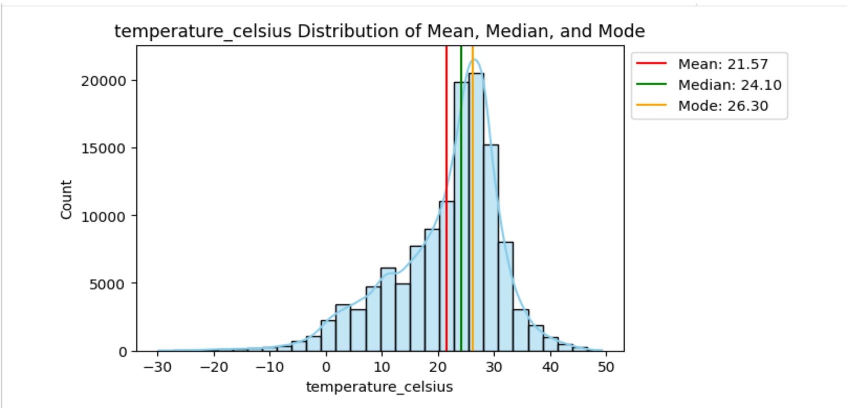


Figure 2: temperature_celsius

Temperature (temperature_celsius): The **mean (21.57°C)** is slightly lower than the **median (24.10°C),** suggesting a **slight left-skewness** due to some lower temperature readings. The **mode (26.30°C)** indicates a frequent occurrence of temperatures around this value.

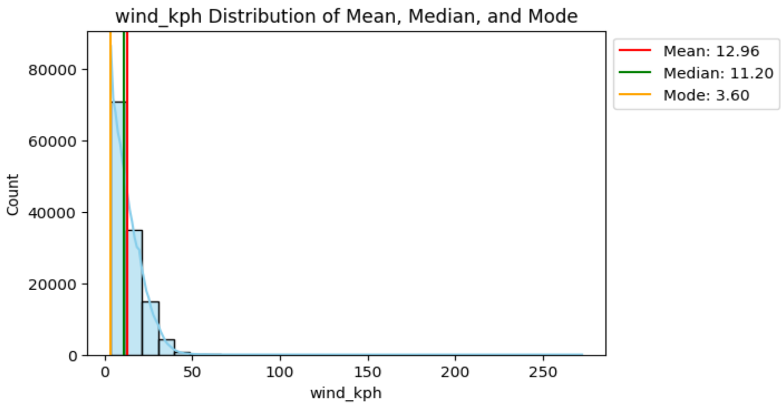


Figure 3: wind_kph

Wind Speed (wind_kph): The **mean (12.96 km/h)** is higher than the **median (11.20 km/h),** implying a **right-skewed distribution**, possibly due to a few instances of higher wind speeds. The **mode (3.60 km/h)** indicates very calm conditions are common.

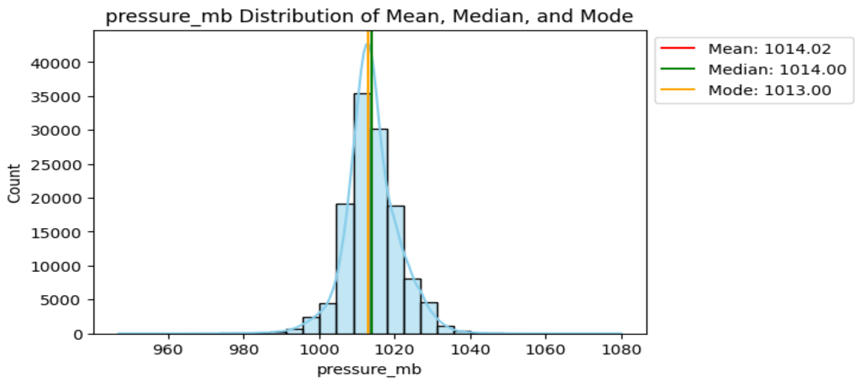


Figure 4: pressure_mb

Atmospheric Pressure (pressure_mb): The **mean (1014.02 mb)** and **median (1014.00 mb)** are very close, indicating a fairly **symmetrical and balanced distribution** of atmospheric pressure.

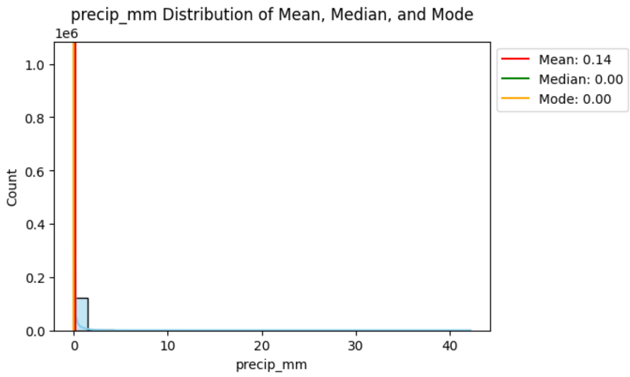


Figure 5: precip_mm

Precipitation (precip_mm): The **mean (0.14 mm**) is significantly higher than both the **median (0.00 mm)** and **mode (0.00 mm).** This **heavily right-skewed distribution** indicates that most observations recorded **no precipitation**, with occasional instances of rainfall.

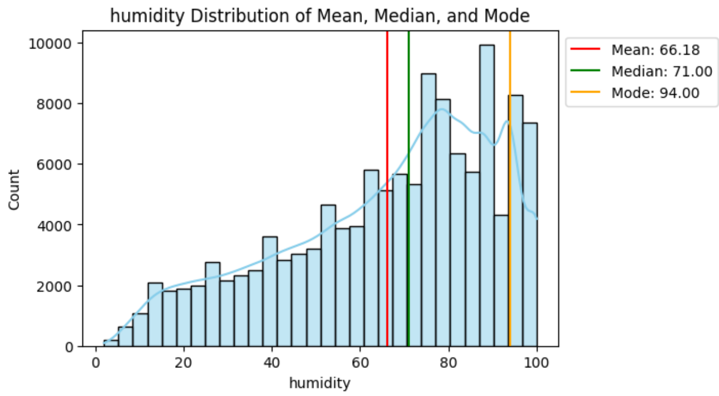


Figure 6: humidity

Humidity: The **mean (66.18%)** is lower than the **median (71.00%),** and the **mode (94.00%)** is considerably higher. This suggests a **left-skewed distribution**, with a large number of observations experiencing high humidity levels.

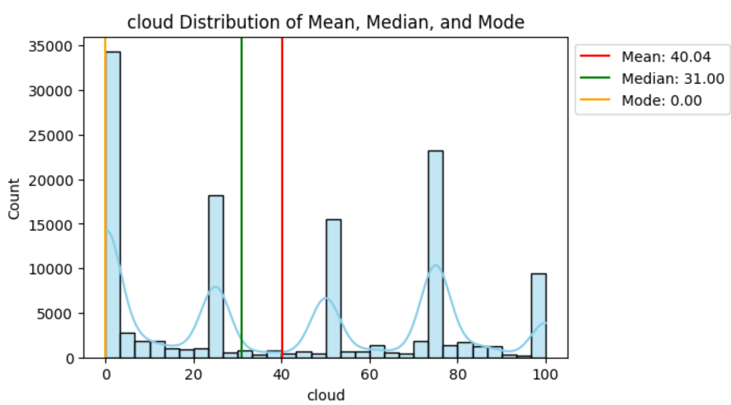


Figure 7: cloud

Cloud Cover: The **mean (40.04%)** is higher than the **median (31.00%),** and the **mode is 0%.** This indicates a **right-skewed distribution**, meaning many observations have clear skies (0% cloud), but some have higher cloud coverage.

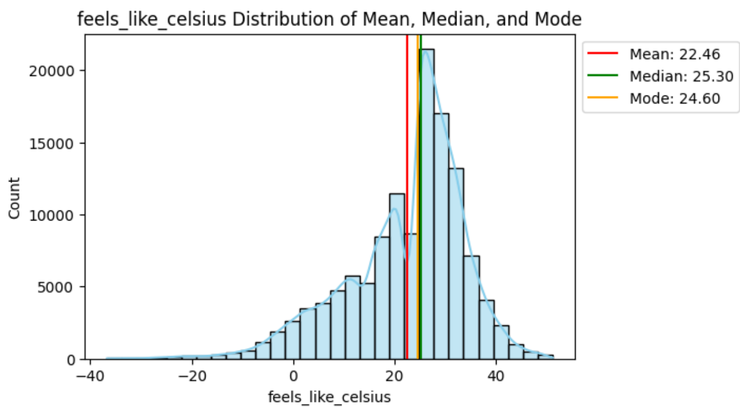


Figure 8: feels_like_celsius

Feels Like Temperature (feels_like_celsius): The **mean (22.46°C**) is lower than the **median (25.30°C),** similar to temperature_celsius, indicating a slight **left-skewness** due to periods of lower perceived temperatures.

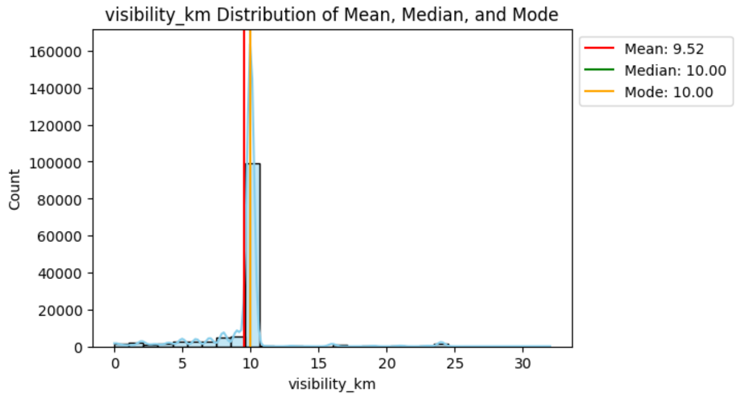


Figure 9: visibility_km

Visibility (visibility_km): The **mean (9.52 km)** is slightly lower than the **median and mode (both 10.00 km),** suggesting that **good visibility (10 km)** is the most common condition, with some instances of reduced visibility.

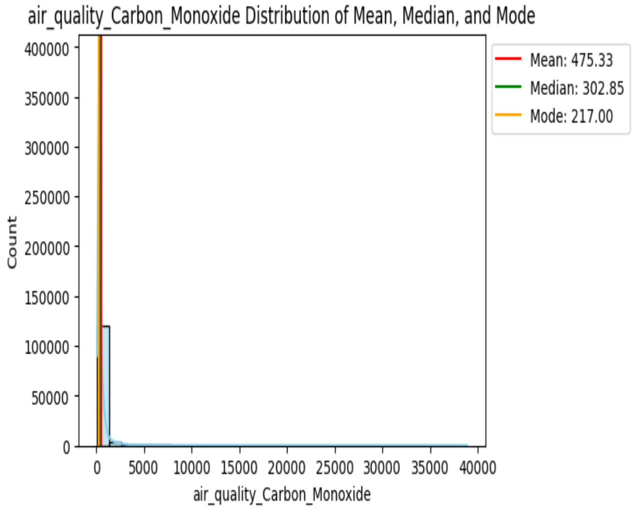


Figure 10: air_quality_carbon_monoxide

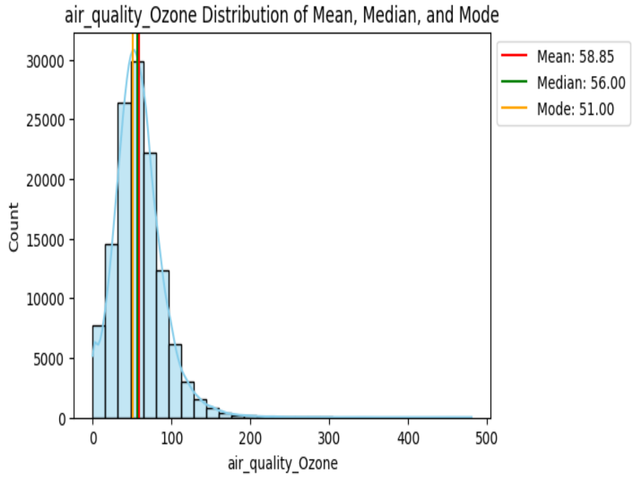


Figure 11: air_quality_Ozone

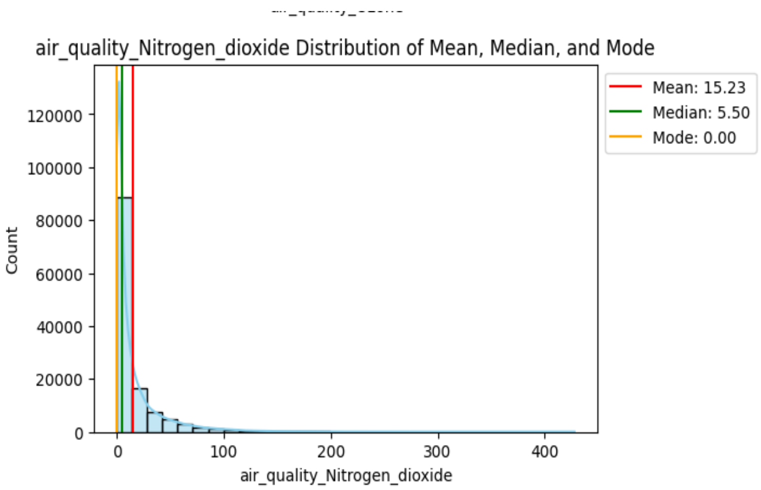


Figure 12: air_quality_Nitrogen_dioxide

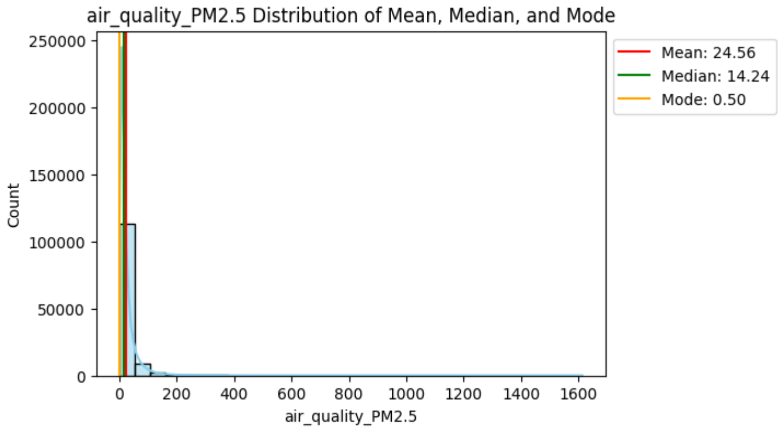


Figure 13: air_quality_PM2.5

Air Quality Parameters (Carbon Monoxide, Ozone, Nitrogen Dioxide, PM2.5, PM10): For most air quality parameters, the mean is significantly higher than the median and mode (where applicable).

This indicates a **strong right-skewness**, meaning that while lower pollution levels are common, there are infrequent but significant occurrences of high pollution concentrations, pulling the mean upwards.

- air_quality_Carbon_Monoxide: Mean (475.33) >> Median (302.85) >> Mode (217.00)
- air_quality_Nitrogen_dioxide: Mean (15.23) >> Median (5.50) >> Mode (0.00)
- air_quality_PM2.5: Mean (24.56) >> Median (14.25) >> Mode (0.50)
- air_quality_PM10: Mean (49.18) >> Median (20.25) >> Mode (12.95)

In summary, the central tendency measures highlight that many variables, especially precipitation and air quality indicators, exhibit right-skewed distributions, implying that lower values are more common, but extreme high values exist and significantly influence the mean.

Measure of Dispersion: Range

The range was calculated for the selected weather and air-quality variables to understand the overall spread and variability of the observations.

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Variables	Min	Max	Range
temperature_celsius	-29.800	49.200	79.000
wind_kph	3.600	272.200	268.600
pressure_mb	947.000	1080.000	133.000
precip_mm	0.000	42.240	42.240
humidity	2.000	100.000	98.000
cloud	0.000	100.000	100.000
feels_like_celsius	-36.700	51.200	87.900
visibility_km	0.000	32.000	32.000
uv_index	0.000	16.300	16.300
gust_kph	3.600	279.400	275.800
air_quality_Carbon_Monoxide	54.850	38879.398	38824.548
air_quality_Ozone	0.000	480.700	480.700
air_quality_Nitrogen_dioxide	0.000	427.700	427.700
air_quality_PM2.5	0.168	1614.100	1613.932
air_quality_PM10	0.168	6037.290	6037.122

- **Temperature** ranges from **-29.8°C to 49.2°C**, showing substantial variation in temperature across different geographical locations.
- **Wind Speed and Gust Speed** have relatively wide ranges of **268.6 km/h** and **275.8 km/h**, respectively, indicating considerable variation in wind conditions.
- **Pressure** has a comparatively narrower range of **133 mb**, suggesting less variation compared with several other numerical variables.
- **Humidity and Cloud Cover** almost their full possible scales, with ranges of **98%** and **100%**, respectively.
- **Precipitation** ranges from **0 to 42.24 mm**, indicating that the dataset contains both dry observations and locations experiencing measurable rainfall.
- **Visibility** ranges from **0 to 32 km**, while the **UV Index** ranges from **0 to 16.3**, showing variation in atmospheric visibility and UV exposure.
- Among the air-quality variables, **CO, PM2.5, and PM10** show very large ranges, indicating substantial variation in pollution levels across the observations.
- The large ranges observed in some variables also indicate the presence of **extreme observations**, which should be considered during further distribution analysis and visualization.

Overall, the range analysis shows that air-quality variables generally exhibit greater variability than atmospheric pressure, while weather variables such as temperature, wind speed, humidity, and cloud cover also show considerable variation across different locations.

Measure Dispersion: Variance

Variance indicates how widely the observations are spread around the mean. A higher variance indicates greater variability, while a lower variance indicates that the observations are relatively closer to the mean.

The variance values obtained for the selected weather and air-quality variables are:

Variables	Variance
temperature_celsius	92.458965
wind_kph	70.863770
pressure_mb	49.211322
precip_mm	0.325355
humidity	577.142071
cloud	1161.979927
feels_like_celsius	132.158749
visibility_km	7.150968
uv_index	12.703395
gust_kph	126.467673
air_quality_Carbon_Monoxide	608187.423778
air_quality_Ozone	966.932293
air_quality_Nitrogen_dioxide	587.048315
air_quality_PM2.5	1421.511537
air_quality_PM10	22925.464586

- **Pressure** has a relatively low variance of **49.21**, indicating that atmospheric pressure values are comparatively stable.
- **Visibility** has a low variance of **7.15**, suggesting that most visibility observations are relatively close to their mean.
- **Precipitation** has a very low variance of **0.33**, indicating that most observations have little or no rainfall, with fewer observations having higher precipitation.
- **Cloud Cover** has a variance of **1161.98**, showing considerable variation in cloud conditions across locations.
- **Temperature** and **Feels-Like Temperature** have variances of **92.46** and **132.16**, respectively, indicating noticeable variation across different locations.
- Among the air-quality variables, **PM10 has the highest variance (22,925.46)**, followed by **Carbon Monoxide (608,187.42)** when considering the raw numerical scale of the variables.
- **PM2.5, Ozone, and Nitrogen Dioxide** also show substantial variability, indicating differences in air-quality conditions across the dataset.

Overall, the variance analysis indicates that **air-quality parameters generally show considerable variability**, while variables such as pressure, visibility, and precipitation show comparatively lower dispersion.

Measure of Dispersion – Standard Deviation

Standard deviation was calculated for the selected weather and air-quality variables to measure how much the observations vary from their respective mean values.

A **low standard deviation** indicates that observations are **closely distributed around the mean**, while a **high standard deviation** indicates greater variability and **wider dispersion from the mean.**

Variables	Standard Deviation
temperature_celsius	9.615559
wind_kph	8.418062
pressure_mb	7.015078
precip_mm	0.570399
humidity	24.023781
cloud	34.087827
feels_like_celsius	11.496032
visibility_km	2.674129
uv_index	3.564182
gust_kph	11.245785
air_quality_Carbon_Monoxide	779.863721
air_quality_Ozone	31.095535
air_quality_Nitrogen_dioxide	24.229080
air_quality_PM2.5	37.702938
air_quality_PM10	151.411573

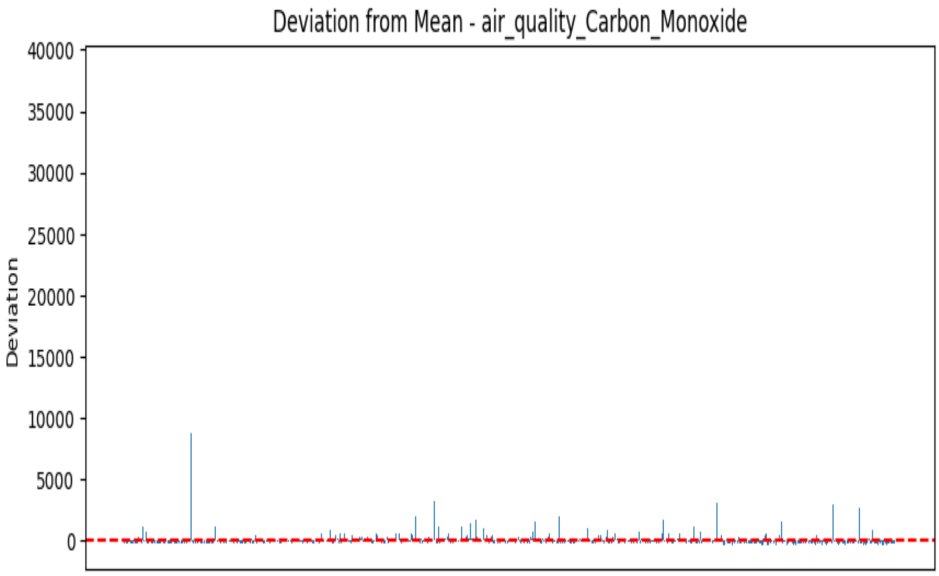


Figure 14: STD - air_quality_Carbon_Monoxide

- The high standard deviation of **779.86** indicates substantial variation in carbon monoxide levels.
- This suggests that CO concentrations vary considerably across different observations and locations.
- The wide spread may also indicate the presence of **extreme pollution values**.

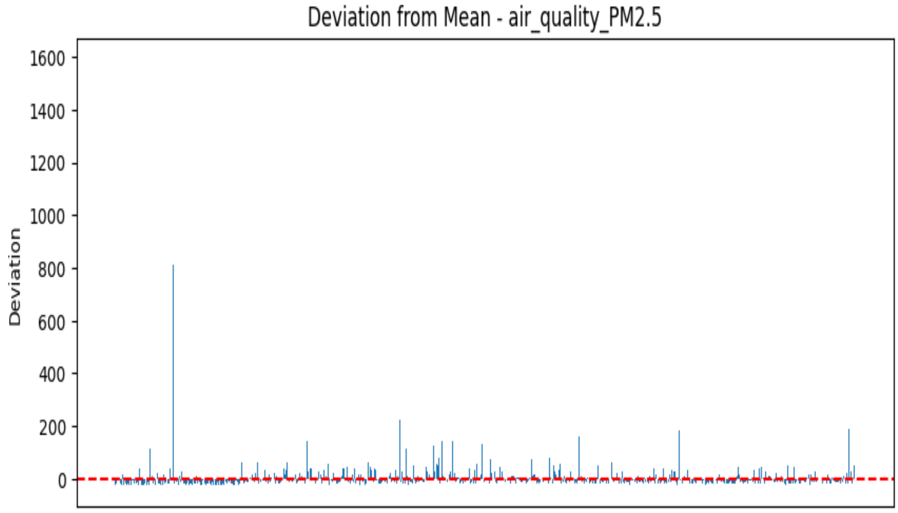


Figure 15: STD – air_quality_PM2.5

- The standard deviation of **37.70** indicates considerable variation in PM2.5 concentrations.
- This suggests that fine particulate matter levels differ noticeably across observations.
- The variation may reflect differences in **pollution conditions across locations**.

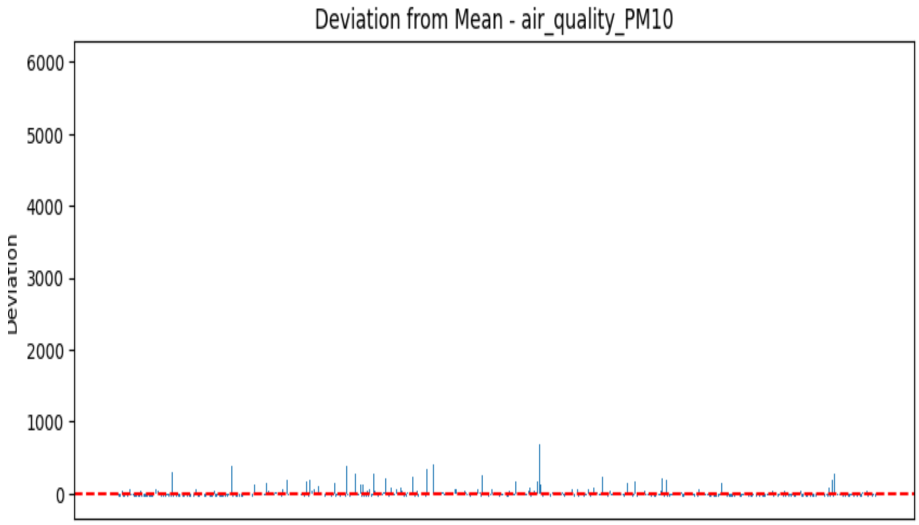


Figure 16: STD – air_quality_PM10

- The standard deviation of **151.41** indicates substantial dispersion in PM10 concentrations.
- This shows that PM10 levels vary considerably across the dataset.
- The high variation may be associated with **different pollution conditions and extreme PM10 observations**.

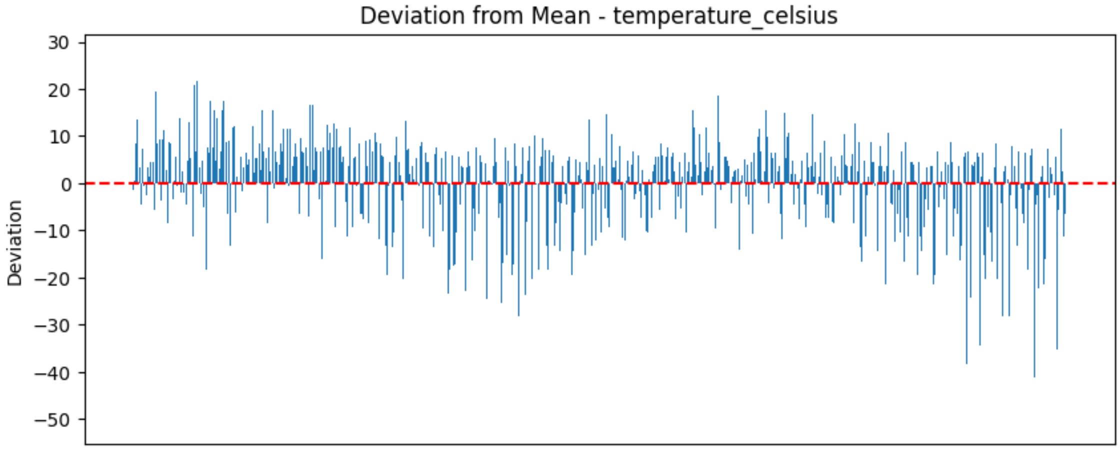


Figure 17: STD – temperature_celsius

- The standard deviation of 9.62°C indicates that temperature values typically vary around the mean by approximately 9.62°C.
- This shows a moderate to relatively high variation in temperature across the observations.
- The variation can be attributed to differences in locations, climates, and weather conditions across the dataset

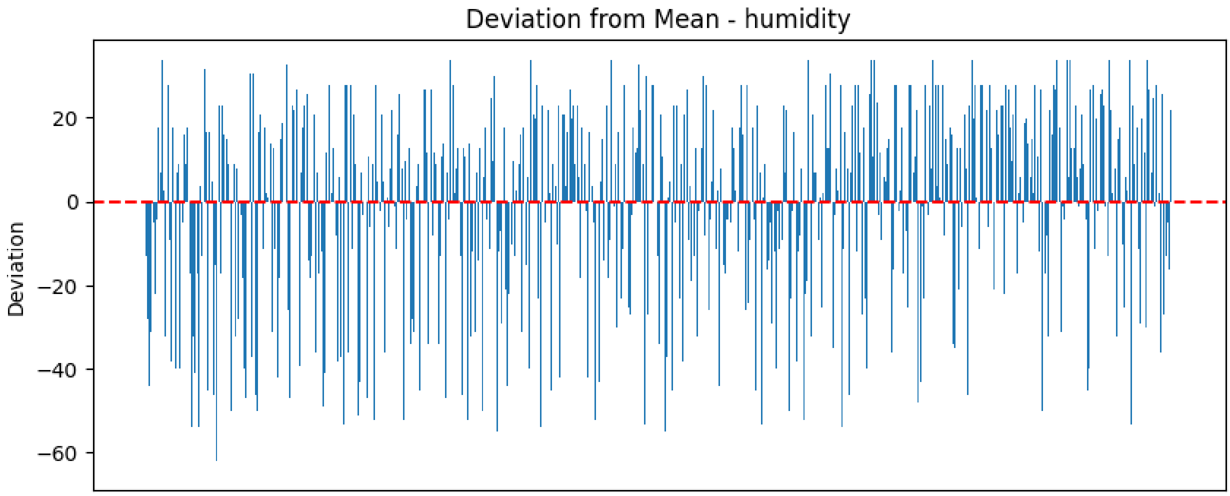


Figure 18: STD - Humidity

- The **standard deviation of 24.02** indicates a **considerable variation** in humidity values around the mean.
- This suggests that humidity levels differ significantly across the observations.
- The variation may be due to differences in locations and weather conditions.

7.7 Data Visualization

Univariate Analysis

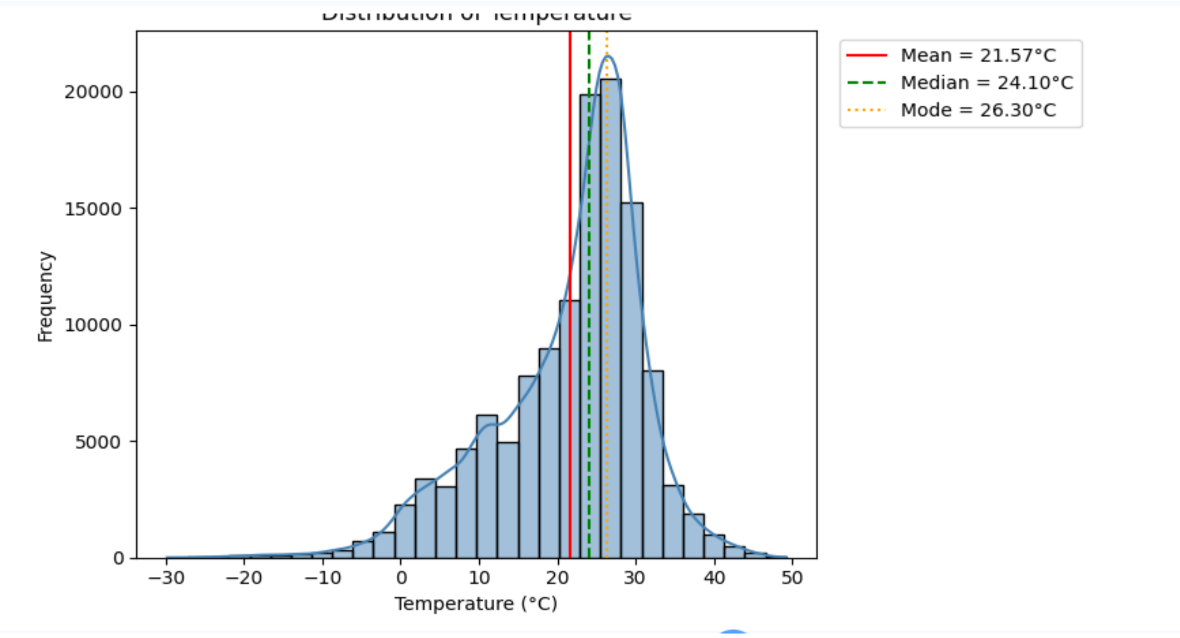
Objective: To analyze the distribution and variability of temperature across the dataset.

Attribute: Temperature_celsius Column

Chart Type: Histogram with Kernal Density Estimation (KDE)

Plot Summary: Analyzes the frequency and density spread of ambient temperatures across dataset.

Visual 1: Temperature Distribution



Key Insights

- The histogram shows a roughly **bell-shaped distribution**, indicating that most temperature readings are concentrated around a central range.
- The highest frequency of observations is between approximately **25°C and 30°C**, suggesting that **moderate-to-warm temperatures** are more common in the dataset.
- The temperature values range from approximately **-30°C to 50°C**, reflecting the dataset's coverage of **diverse climatic** conditions across different countries and locations.
- **Extreme cold and hot** temperatures occur less frequently, indicating that most observations fall within a moderate temperature range.
- Overall, the distribution suggests that typical **global temperature** observations are concentrated around **moderate-to-warm conditions**, while extreme temperatures are relatively uncommon.

Most Frequent Weather Conditions

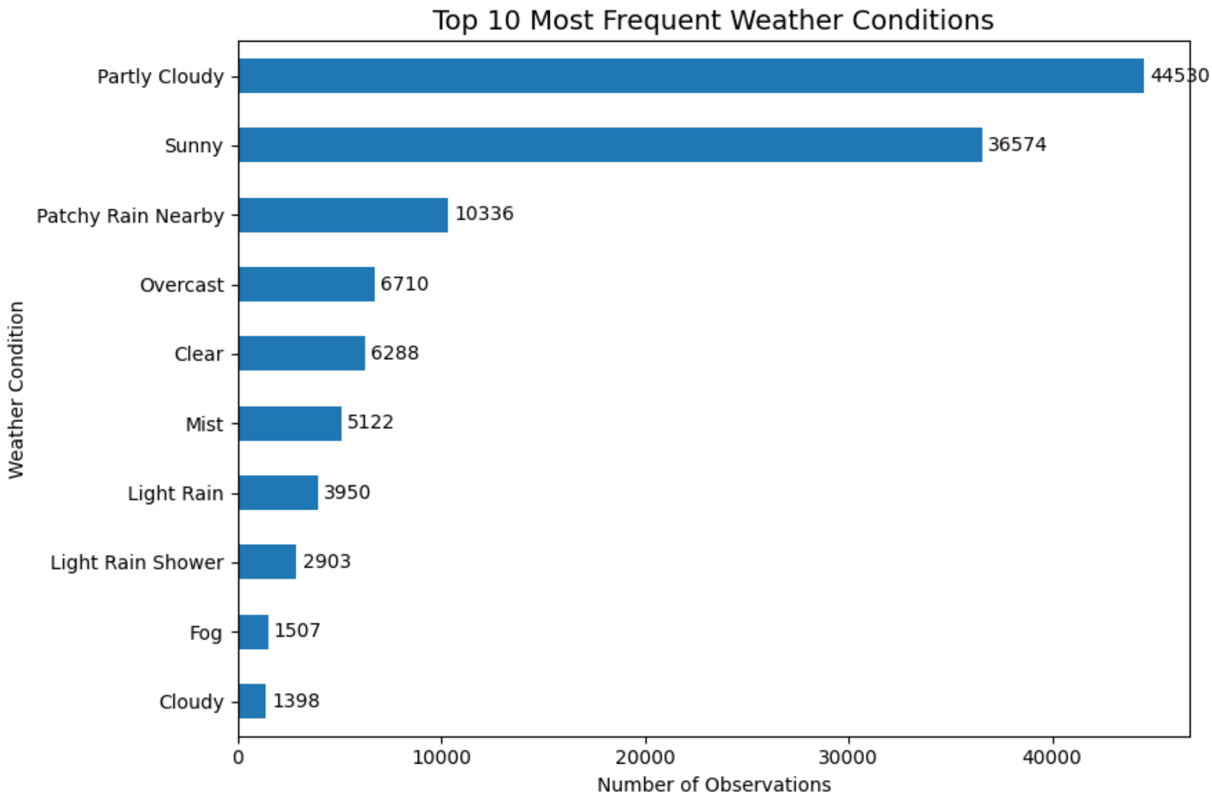
Objective: To identify and visualize the top most frequently occurring weather conditions in the dataset

Attribute: Weather_Condition Column

Chart Type: Horizontal Bar Chart

Plot Summary: To determine the most common weather conditions and compare their frequency using the top 10 weather conditions.

Visual 2: Most Frequent Weather Conditions



Key Insights

- **Partly Cloudy** is the most frequently recorded weather condition, followed by **Sunny**, indicating that clear or mostly clear conditions dominate the dataset.
- **Light Rain, Clear, and Mist** are also among the common weather conditions, but occur less frequently than **Sunny and Partly Cloudy.**
- There is a noticeable **drop in frequency after the top few conditions**, showing that some weather conditions are relatively uncommon.
- Conditions such as **Moderate or Heavy Rain with Thunder** and **Freezing Fog** occur much less frequently, indicating that severe or unusual weather events are relatively rare in the dataset.
- Overall, the distribution shows that the dataset is **dominated by common and relatively mild weather conditions** rather than extreme weather events.

Bivariate Analysis

Weather Conditions across Countries

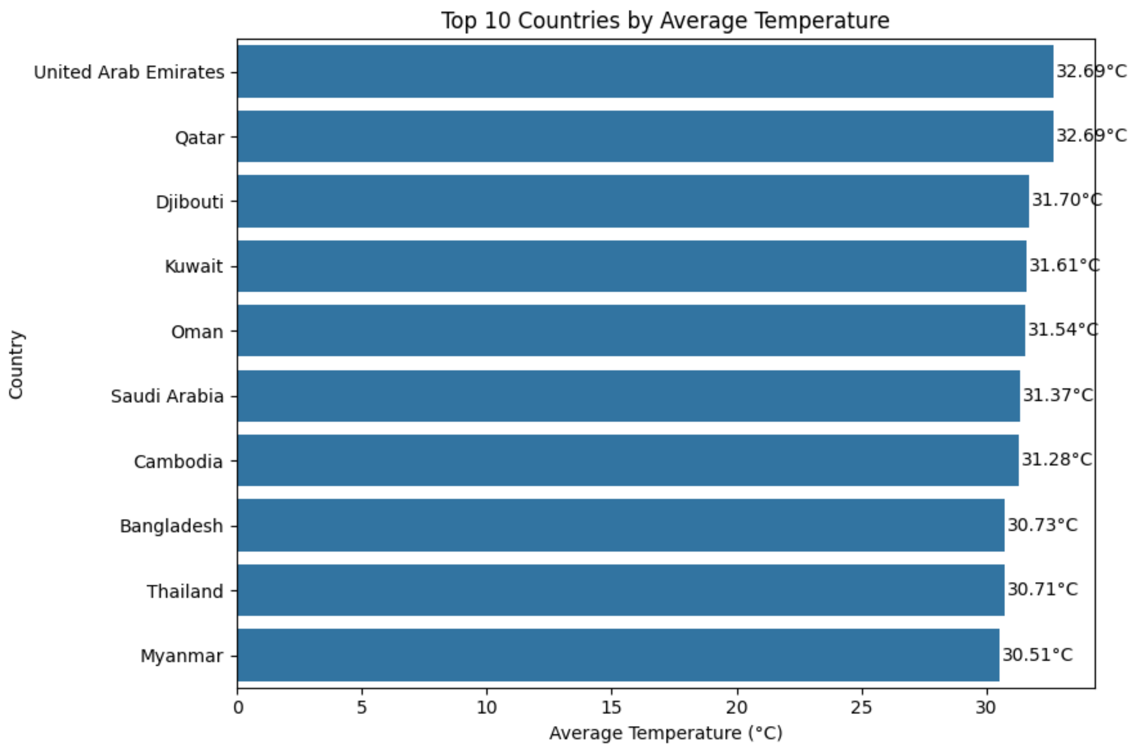
Objective: Analyze global weather conditions across countries

Attributes: Country & Temperature_Celsius Column

Chart Type: Horizontal Bar Chart

Plot Summary: Compare the average temperature across the top 10 countries.

Visual 3: Top 10 Countries by Average Temperature



Key insights

- United Arab Emirates has the highest average temperature at approximately 32.69°C, closely followed by Qatar (32.69°C).
- Djibouti, Kuwait, Oman, and Saudi Arabia also show high average temperatures above 31°C, indicating predominantly hot climatic conditions.
- Cambodia, Bangladesh, Thailand, and Myanmar have average temperatures above 30°C, showing that several Southeast Asian countries also experience consistently warm conditions.
- The difference between the highest and lowest average temperatures among these top 10 countries is only about 2.18°C, indicating that the countries in this group have relatively similar warm temperature levels.
- Overall, the top 10 countries are concentrated mainly in hot and tropical regions, where warm temperatures are commonly observed.

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Relationship between Humidity & Air Quality PM2.5

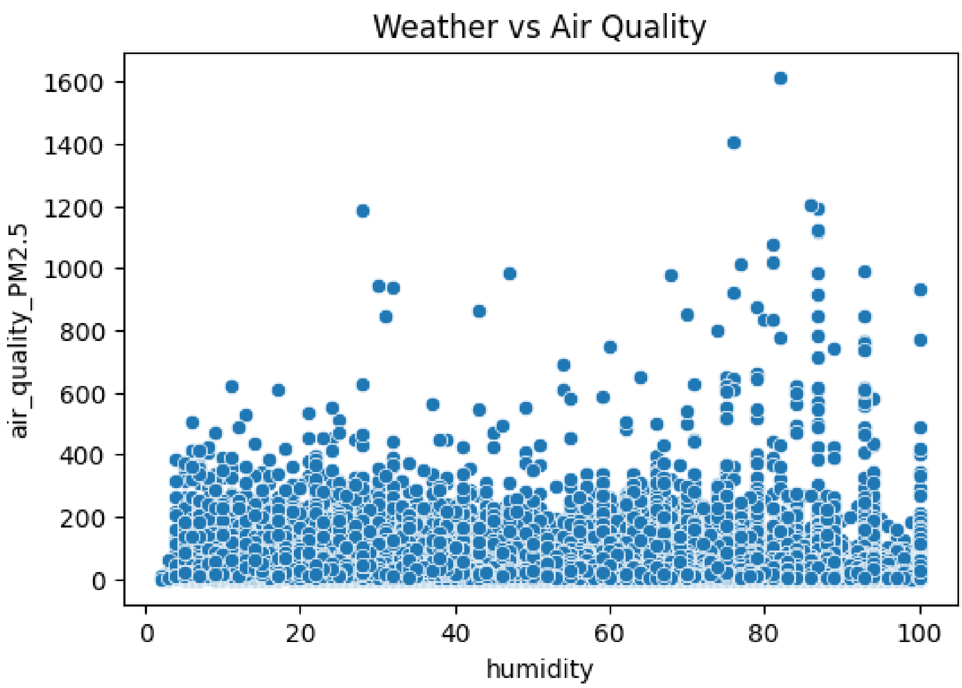
Objective: To identify relationships between weather parameters and air quality indicators.

Attribute: Humidity & Air_Quality_PM2.5 Column

Chart Type: Scatter Plot

Plot Summary: To determine whether changes in humidity are associated with changes in PM2.5 concentration.

Visual 4: Weather vs Air Quality



Key Insights

- Most observations show **low to moderate PM2.5 levels across different humidity values**.
- There is **no strong clear linear relationship** between humidity and PM2.5 in this chart.
- Some **high PM2.5 values occur at higher humidity levels**, particularly around **70–90% humidity**, but **they are relatively few.**
- The wide spread of PM2.5 values indicates that **humidity alone may not explain pollution levels**; other factors such as wind speed, temperature, and location may also influence air quality.
- A few **extreme PM2.5 values** appear as outliers, reaching above **1,000 µg/m³**.

Actual Temperature vs Feels like Temperature

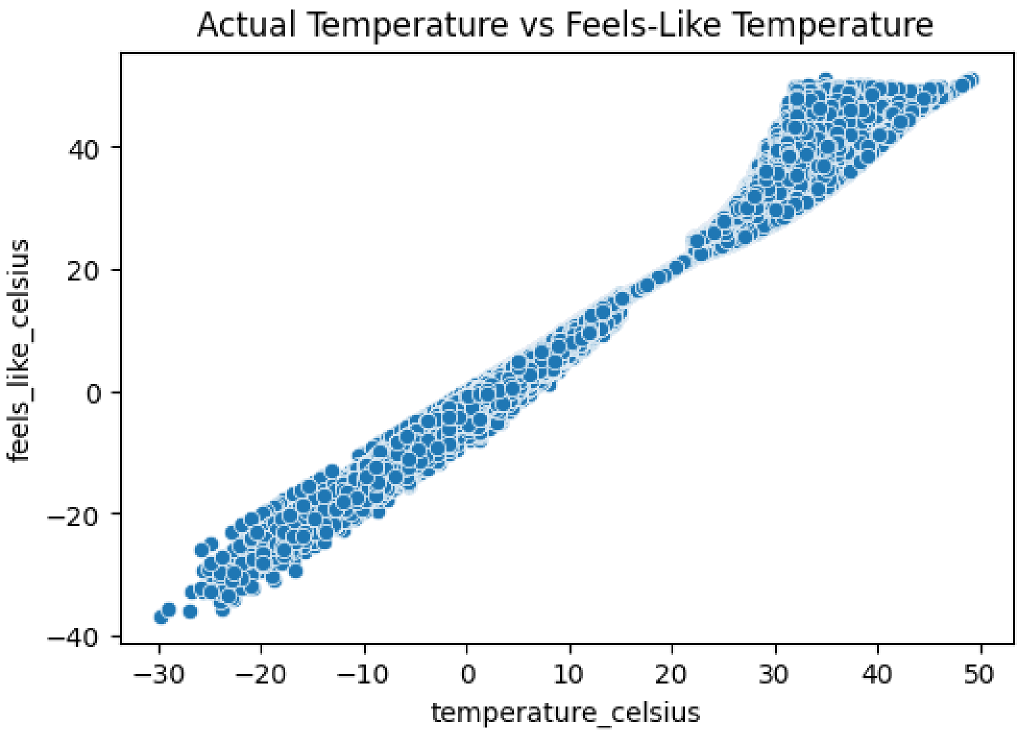
Objective: To analyze the relationship between actual temperature and feels-like temperature

Attributes: Temperature_Celsius & Feels_like_celsius

Chart Type: Scatter Plot

Plot Summary: To analyze the relationship between actual temperature and feels-like temperature and understand how closely they vary across different weather conditions.

Visual 5: Actual Temperature vs Feels like temperature



- The scatter plot shows a **strong positive relationship** between actual temperature and feels-like temperature.
- As **actual temperature increases, the feels-like temperature also generally increases**.
- At lower temperatures, the two values are relatively close, while at higher temperatures, there is **greater variation** in feels-like temperature.
- This variation suggests that **humidity, wind speed, and other weather factors can influence how the temperature feels**.
- Overall, the plot indicates that **actual temperature is a strong indicator of feels-like temperature**.

Identify Higher Pollution Countries

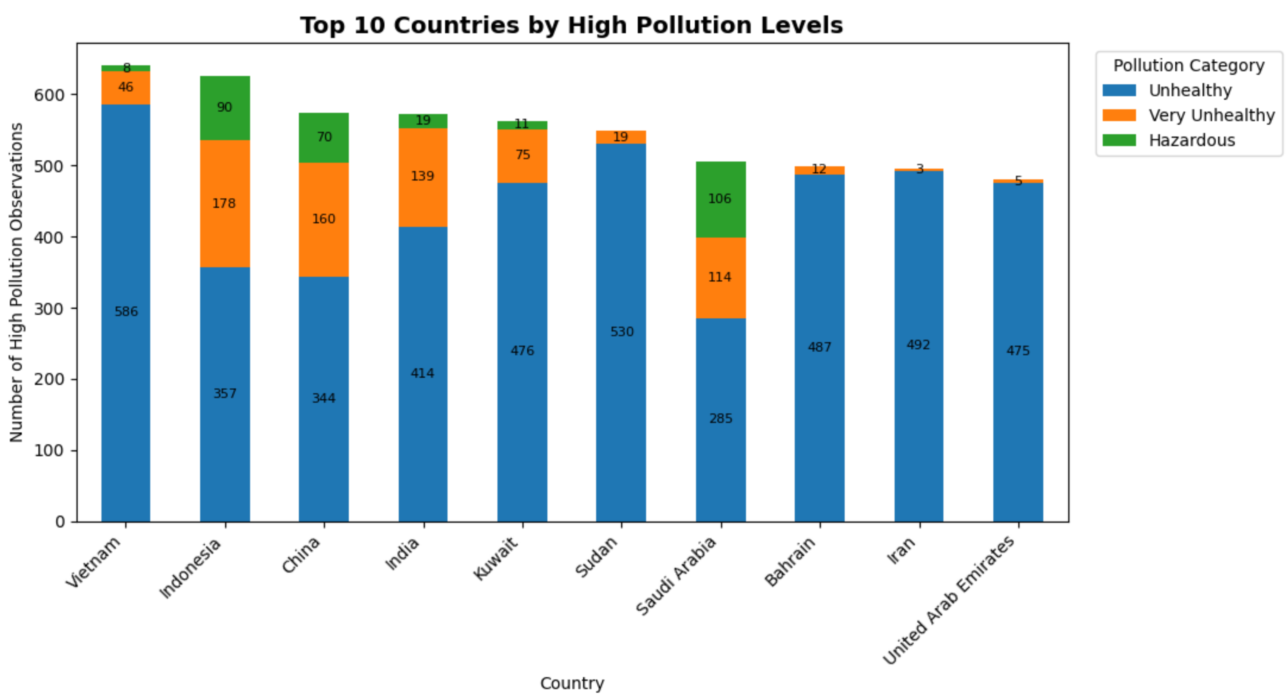
Objective: To identify locations with higher pollution levels using air quality indicators and pollution category.

Attributes: Country & Pollution_Category Columns

Chart Type: Stacked Bar Chart

Plot Summary: The stacked bar chart identifies the top 10 countries with the highest high-pollution observations and shows the contribution of each pollution severity category.

Visual 6: Higher Pollution Country



Key Insights

- **Vietnam** has the highest number of high-pollution observations, with **586 Unhealthy, 46 Very Unhealthy, and 8 Hazardous** observations.
- **Indonesia** has the highest number of **Very Unhealthy (178)** observations among the countries shown.
- **Saudi Arabia** has a notable number of **Hazardous observations (106)**, the highest in this chart.
- **China and India** also show substantial high-pollution observations, with **Unhealthy** being the dominant category.
- Overall, **Unhealthy pollution is the most common high-pollution category** across most of the top 10 countries, while **Hazardous** observations are generally less frequent.
- **Iran and the United Arab Emirates** have comparatively fewer Very Unhealthy and Hazardous observations than several other countries in the chart.

Monthly Distribution of Pollution Category

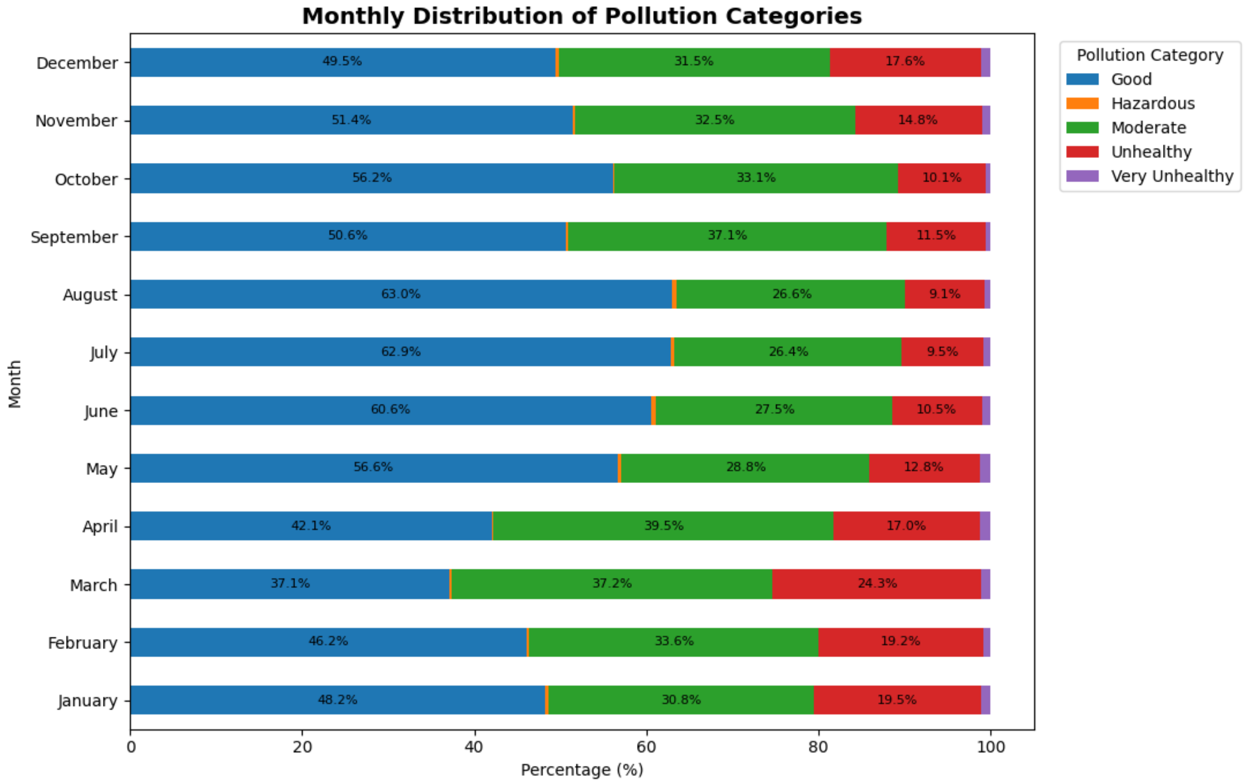
Objective: To analyze the monthly distribution and variation of pollution categories.

Attributes: Country and Month_name Column

Chart Type: Horizontal Bar Chart

Plot Summary: The chart highlights monthly variations in pollution categories and identifies months with higher proportions of unhealthy air-quality levels.

Visual 7: Month wise Pollution Categories



Key Insights:

- **Good** pollution category is highest in **August (63.0%) and July (62.9%)**, indicating better air-quality conditions during these months.
- **March** shows the **lowest Good category (37.1%)** and the **highest Unhealthy category (24.3%)**, indicating relatively poorer air quality.
- **April has the highest Moderate category (39.5%)**, followed by March (37.2%) and September (37.1%).
- **January and February** also show relatively higher Unhealthy proportions at **19.5% and 19.2%**, respectively.
- **Very Unhealthy and Hazardous** categories represent only a **small proportion** of observations across most months.
- Overall, the chart suggests **better pollution conditions during June–August and comparatively poorer conditions during March–April and January–February**.

Multi Variate Analysis

Pollution based Correlation Analysis

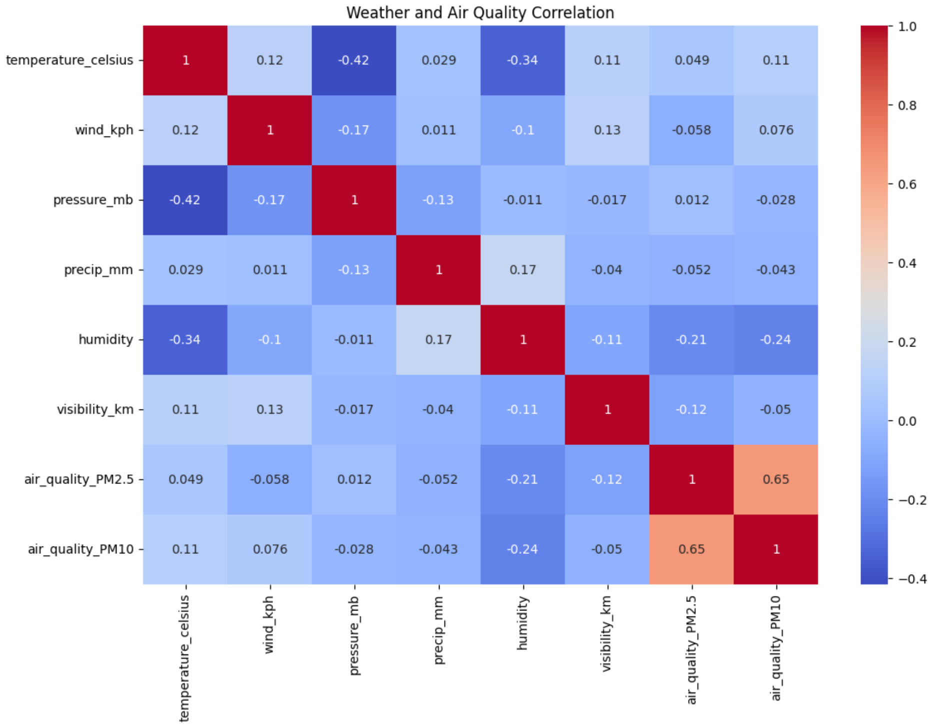
Objective: To identify relationships between weather parameters and air quality.

Attributes: temperature_celsius, wind_kph, pressure_mb, precip_mm, humidity, visibility_km, air_quality_PM2.5, air_quality_PM10

Chart Type: Correlation Heatmap

Plot Summary: To analyze the correlation between weather parameters and air quality indicators and identify the strength and direction of their relationships.

Visual 8: Pollution based Correlation Analysis



Key Insights

- **PM2.5 and PM10**: These two air quality indicators are **highly correlated (0.97),** which is expected as they are both particulate matters. This means they **tend to increase or decrease together.**
- **Temperature and Pressure**: There's a moderate **negative correlation (-0.415),** suggesting that as **temperature rises, atmospheric pressure generally tends to fall.**
- **Humidity and Particulate Matter**: Humidity shows a moderate **positive correlation with PM2.5 (0.33) and PM10 (0.39).** This implies that higher humidity levels might be associated with increased concentrations of particulate matter.
- **Visibility and Particulate Matter**: Visibility has a **negative correlation with PM2.5 (-0.23) and PM10 (-0.28).** This indicates that poorer air quality (higher PM2.5/PM10) leads to reduced visibility.
- **Wind Speed and Precipitation:** Wind speed (wind_kph) and precipitation (precip_mm) show relatively **weak linear correlations with air quality indicators** in this dataset, suggesting their direct linear relationship with pollution levels might not be strong or is more complex.

Temperature vs PM2.5 by Pollution Category

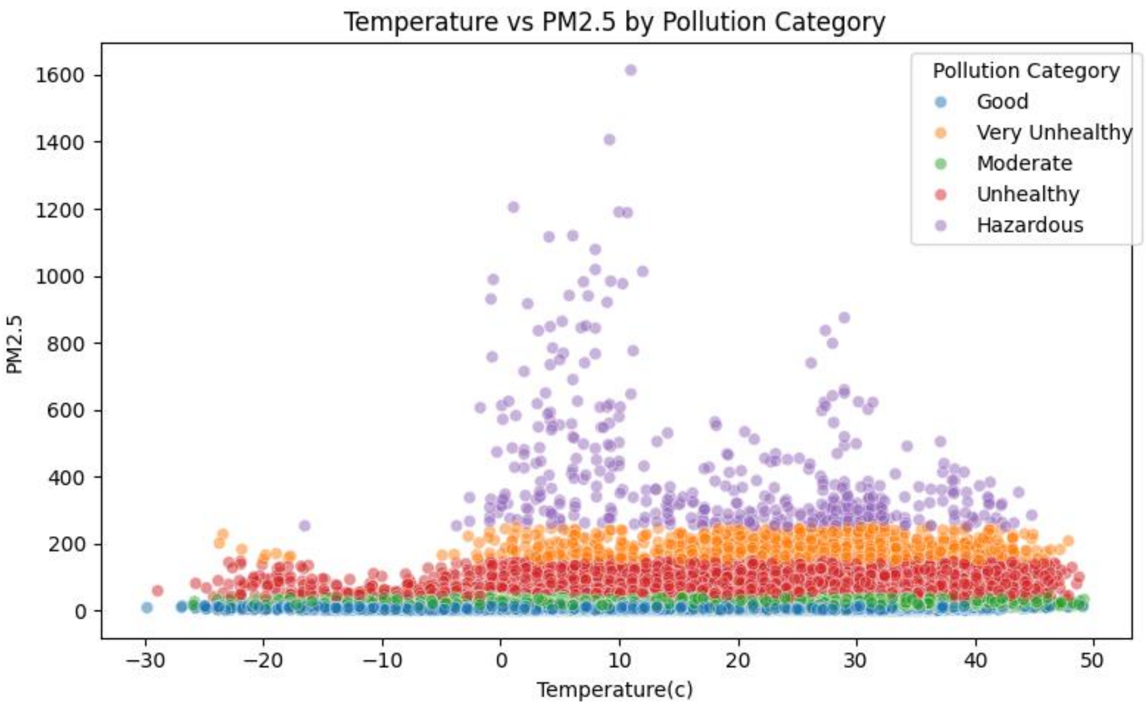
Objective: To analyze the relationship between temperature and PM2.5 levels across different pollution categories.

Attributes: temperatue_celsius, air_quality_PM2.5 and Pollution_Category

Chart Type: Scatter Plot

Plot Summary: The scatter plot illustrates the relationship between temperature and PM2.5 levels across different pollution categories, highlighting variations in air quality at different temperature ranges.

Visual 9: Temperature vs PM2.5 Across Pollution Categories



Key Insights

- **Good** pollution observations are mainly concentrated at **lower PM2.5 levels** across most temperature ranges.
- As **PM2.5 increases**, observations shift toward **Moderate, Unhealthy, Very Unhealthy, and Hazardous** categories.
- The highest PM2.5 values are mostly observed around **0–15°C**, with some extreme values exceeding **1,000 µg/m³**.
- At moderate-to-high temperatures (**20–40°C**), PM2.5 values are generally lower than the extreme concentrations seen around 0–15°C.
- Overall, the plot shows **variation in PM2.5 across temperature ranges**, but temperature alone does not appear to consistently determine pollution levels.

Hourly Variation in PM2.5 & PM10 Levels

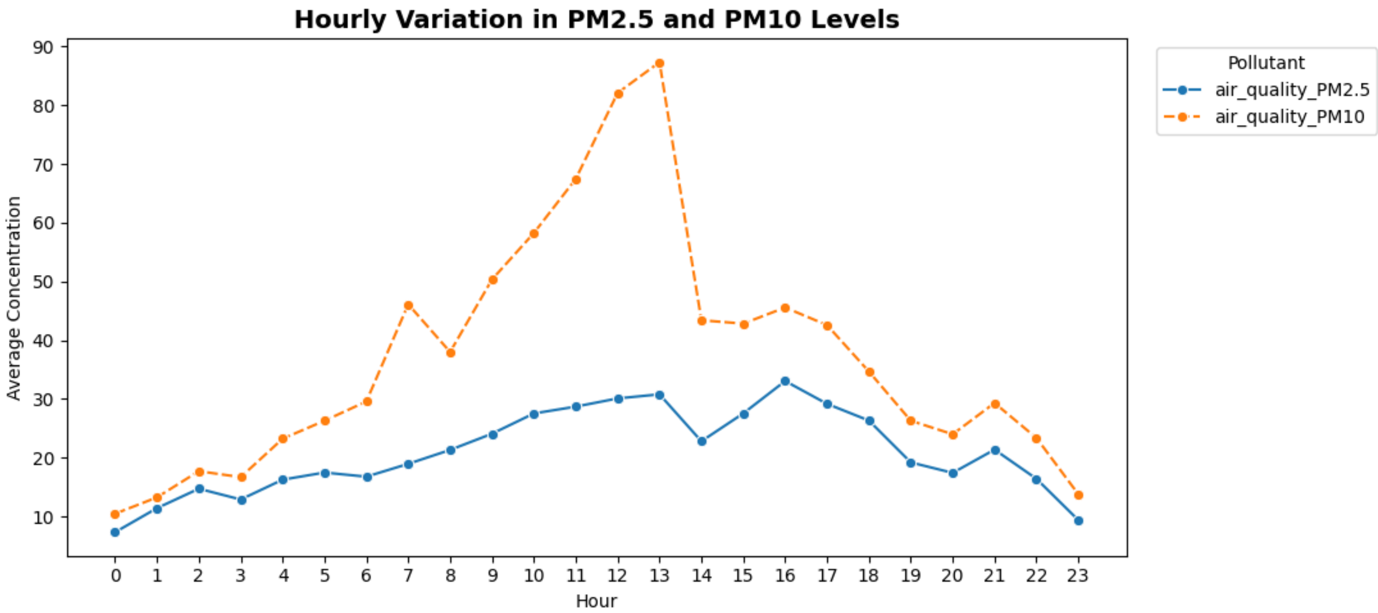
Objective: To identify hourly patterns in PM2.5 and PM10 pollution levels.

Attribute: update_hour, air_quality_PM2.5 and air_quality_PM10

Chart Type: Line Chart

Plot Summary: The line chart highlights hourly variations in PM2.5 and PM10 levels, helping identify periods of higher and lower pollution throughout the day.

Visual 10: Hourly pattern in PM2.5 & PM10



Key Insights

Early Morning Peak (around 0-3 AM): There's an initial peak in both PM2.5 and PM10 concentrations during the early morning hours, particularly around 1-3 AM.

Morning Decline: Following the early morning peak, pollution levels tend to decrease, reaching their lowest point in the late morning to early afternoon (roughly 6 AM to 12 PM).

Afternoon/Evening Increase: Both PM2.5 and PM10 levels begin to rise again in the afternoon, continuing into the evening.

Evening Peak (around 6-9 PM): Another significant peak in pollution is observed in the evening hours, typically between 6 PM and 9 PM.

PM10 generally higher than PM2.5: Throughout the day, PM10 concentrations are consistently higher than PM2.5 concentrations.

These patterns often correlate with human activity, such as morning and evening commutes, and atmospheric conditions like temperature inversions that can trap pollutants closer to the ground during specific times of the day.

8.Four Types of Analysis

📊 Descriptive Analysis

- The dataset covers weather and air-quality observations across multiple countries and cities, providing a global view of environmental conditions.
- Temperature observations are mainly concentrated around the 15°C–32°C range, while extreme temperatures occur less frequently.
- The analysis shows that Good and Moderate air-quality conditions are common, but unhealthy pollution categories are also present in several locations.
- PM10 is generally higher than PM2.5, indicating the presence of both fine and coarse particulate pollution.
- Country-wise analysis shows significant variation in average PM2.5 and PM10 concentrations.
- Higher PM2.5 concentrations are observed in some countries, indicating locations with comparatively poorer air quality.
- Wind speed and humidity show noticeable variation across observations, while extreme wind-speed values are relatively uncommon.

🔍 Diagnostic Analysis

- The Temperature vs PM2.5 analysis shows that PM2.5 occurs across a wide range of temperatures, suggesting that temperature alone does not explain pollution levels.
- The Wind Speed vs PM2.5 visualization indicates a possible negative relationship, where higher PM2.5 concentrations are more frequently observed at lower wind speeds.
- The Humidity vs PM2.5 analysis shows a possible positive relationship, with higher PM2.5 values occurring more frequently at higher humidity levels.
- Pollution categories show clear differences in PM2.5 concentration: Good and Moderate categories generally have lower PM2.5, while Unhealthy, Very Unhealthy and Hazardous categories have progressively higher values.
- Country-wise PM2.5 and PM10 averages indicate that geographical location is an an important factor in pollution variation.
- Extreme values in variables such as wind speed and pollutant concentrations were identified as potential data errors or outliers and were reviewed during data cleaning.

💡 **Prescriptive Analysis**

- Locations showing consistently high PM2.5 and PM10 should be prioritized for air-quality monitoring and pollution-control measures.
- During periods of high pollution and low wind conditions, authorities can issue health advisories and recommend reducing outdoor exposure.
- Areas experiencing recurring unhealthy pollution levels can implement stronger emission-control strategies targeting traffic, industries and other pollution sources.
- Real-time monitoring can be used to provide early warnings when air quality approaches unhealthy levels.
- Weather conditions associated with higher pollution can be incorporated into an early-warning system for vulnerable populations.
- Countries and cities with comparatively cleaner air can be studied to identify better environmental practices.

🔮 **Predictive Analysis**

- Weather variables such as temperature, humidity, wind speed, cloud cover and visibility, together with pollutant measurements, can be used to predict future air-quality conditions.
- PM2.5 and PM10 levels can be used as important indicators for identifying the likelihood of higher pollution categories.
- Locations that repeatedly show high PM2.5/PM10 concentrations may have a higher likelihood of experiencing poor air-quality conditions.
- Historical weather and air-quality patterns can potentially be used to predict whether a location may fall into Good, Moderate, Unhealthy or Hazardous pollution categories.
- A machine-learning classification model could be developed to predict the US EPA pollution category using weather and pollutant features.

10.Future Enhancement

Pollution Monitoring: Locations with higher PM2.5 and PM10 levels should be monitored regularly to identify areas experiencing poor air quality.

Real-Time Monitoring: Real-time weather and air-quality data can be integrated to provide continuous pollution monitoring.

Pollution Forecasting: Machine learning models can be developed to predict future PM2.5 and PM10 levels and support early pollution warnings.

Weather-Based Analysis: Weather conditions such as temperature, humidity and wind speed should be considered when assessing changes in pollution levels.

Pollution Alerts: Automated alerts can be implemented when pollution levels reach unhealthy or hazardous categories.

Geographical Analysis: Pollution hotspots can be identified using country, city, latitude and longitude information to support targeted environmental actions.

11.Conclusion

The Global Weather & Air Quality Analysis provides valuable insights into weather conditions and air-quality levels across different countries and cities.

The analysis identified significant variations in temperature, humidity, wind speed, PM2.5, and PM10 concentrations across locations. The results also indicate that factors such as wind speed, humidity, temperature, and geographical location may influence air-quality conditions.

Some locations showed higher particulate pollution and unhealthy air-quality categories, highlighting potential areas of concern.

Overall, this analysis can support better air-quality monitoring, pollution assessment, early-warning systems, and data-driven environmental decision-making.